

Tunable Isolator-Free Magnetron Power-Combining System Based on Peer-to-Peer Locking Technique With a Three-Stub Tuner

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Abstract—Magnetrons are widely used microwave sources in industrial applications. To overcome the power limitations of a single magnetron and achieve large-power output, a compact, high efficiency, and straightforward magnetron power-combining system based on peer-to-peer injection locking is proposed. An *H*-plane tee integrated with a three-stub tuner is employed as a coupling network between two magnetrons, with the coupling characteristics adjusted via the insertion depths of the stubs. The injection-locking characteristic is qualitatively analyzed and experimentally verified. Without circulators or external injection sources, the system based on two commercial magnetrons achieves the phase synchronization through peer-to-peer injection locking. The proposed system achieves a maximum microwave efficiency of 90.1% at an output power of 1440 W. It maintains 87.8% efficiency at a peak output power of 1750 W. This work provides a novel perspective on the development of coherent magnetron power-combining systems based on peer-to-peer injection-locked magnetrons.

Index Terms—High efficiency, injection locking, magnetron, peer-to-peer locking, power combining.

I. INTRODUCTION

MICROWAVE applications, such as microwave heating, microwave chemistry, microwave wireless power transmission (MWPT), and so on, have received increasing interest and developments in recent years [1], [2], [3], [4], [5]. Magnetrons have become preferred microwave sources for industrial applications because of their advantages of high efficiency, low cost, and compact size. However, a magnetron's output power is limited by its resonant cavity size [6].

Received 7 January 2026; accepted 28 January 2026. Date of publication 26 February 2026; date of current version 8 July 2026. This work was supported in part by the National Natural Science Foundation of China (NSFC) through JCKY under Grant U22A2015 and Grant 60301004; in part by China Scholarship Council under Grant 202406240217; and in part by the Collaborative Research Program of Microwave Energy Transmission Laboratory (METLAB), Research Institute for Sustainable Humanosphere, Kyoto University. (Corresponding author: Changjun Liu.)

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Digital Object Identifier 10.1109/LMWT.2026.3661498

Therefore, coherent power combining, as a technology that overcomes a magnetron's power limit and satisfies the demand for large-power output, has attracted significant attention from researchers.

Injection-locking techniques have been widely adopted to control the frequency and phase of magnetrons and achieve high-efficiency coherent power combining [7], [8], [9], [10], [11], [12]. However, a magnetron power-combining system typically requires circulators and phase shifters, which significantly increase the cost and make the system complex.

Peer-to-peer injection-locking has attracted great interest in recent years. First, Pengvanich et al. [13] conducted a theoretical analysis of a general model for peer-to-peer injection-locking, which was later experimentally verified by Cruz et al. [14]. Cheng et al. [15] designed a novel coupling bridge of magnetrons and achieved peer-to-peer locking with a relative phase jitter less than $\pm 1^\circ$. Li et al. [16] combined external injection with the mutual injection to achieve phase control of two magnetrons. Hou et al. [17] utilized a waveguide coupler to realize peer-to-peer locking among magnetrons and successfully achieved a magnetron array. The characteristic of low cost and compact size of a peer-to-peer locking system makes it well-suited for the construction of a power-combining system.

In our previous work [18], we demonstrated that the inherent coupling characteristic of an *H*-plane tee enables peer-to-peer injection-locking between two magnetrons. In this article, a three-stub tuner is employed in a peer-to-peer locked magnetron power-combining (PPL-MPC) system to adjust the phase characteristics of the system. We experimentally investigate the influence of the three-stub tuner on the system's coupling behavior and phase response. Meanwhile, the MPC system is also conducted using two commercial S-band 1-kW magnetrons. The system achieves a power-combining efficiency of 90.1%. This method provides a novel idea for the design of an MPC system.

II. PPL-MPC SYSTEM WITH A THREE-STUB TUNER

A. Analysis of a Three-Stub Tuner in the Proposed System

A three-stub tuner is a commonly used waveguide impedance-matching component that consists of three independently adjustable screws, each of which can be adjusted in its insertion depth. The three-stub tuner behaves like a

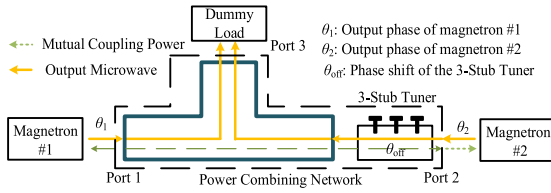


Fig. 1. Diagram of the proposed coherent power-combining system.

shunt susceptance of jB and introduces extra phase shift and amplitude variation in a waveguide system.

The three-stub tuner is equivalent to a two-port network. The corresponding susceptance jB is controlled by adjusting the insertion depth of each tuning stub, thereby enabling impedance tuning of the network and modifying its S -parameters. We assume that the system has a characteristic impedance of Z_0 and defined the introduced phase shift as θ_{off} . Its S -parameter is

$$[S_{3\text{-stub}}] = \begin{bmatrix} \frac{-jBZ_0}{2 + jBZ_0} & \frac{2}{2 + jBZ_0} \\ \frac{2}{2 + jBZ_0} & \frac{-jBZ_0}{2 + jBZ_0} \end{bmatrix}. \quad (1)$$

The microwave amplitude variation $A_{3\text{-stub}}$ and the phase variation θ_{off} are derived as follows:

$$A_{3\text{-stub}} = |S_{21}| = \frac{2}{\sqrt{4 + (BZ_0)^2}} \quad (2)$$

$$\theta_{\text{off}} = \angle S_{21} = -\arctan\left(\frac{BZ_0}{2}\right). \quad (3)$$

Therefore, using the jB of the three-stub tuner, simultaneous modulation of microwave amplitude and phase is achieved. Based on this characteristic, we propose a phase-shifter-free and isolator-free injection-locked magnetron power-combining system using a three-stub tuner, as shown in Fig. 1. The three-stub tuner is placed between a tee and a magnetron to shift the phase for magnetrons' peer-to-peer locking.

B. Peer-to-Peer Locking of Magnetrons

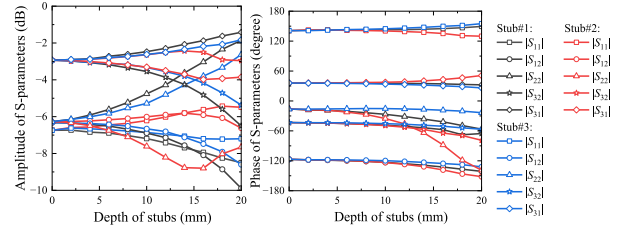
The proposed PPL-MPC system is illustrated in Fig. 1. Two magnetrons are connected to the two input ports of the power-combining network, which consists of a symmetric H -plane tee and a three-stub tuner. The S -parameter matrix of the overall power-combining network is

$$[S_{\text{PCN}}] = \begin{bmatrix} |S_{11}|e^{j\alpha_1} & |S_{12}|e^{j\beta} & |S_{13}|e^{j\gamma} \\ |S_{21}|e^{j\beta} & |S_{22}|e^{j\alpha_2} & |S_{23}|e^{j\sigma} \\ |S_{31}|e^{j\gamma} & |S_{32}|e^{j\sigma} & |S_{33}|e^{j\alpha_3} \end{bmatrix}. \quad (4)$$

Among them, $|S_{nm}|$ and $\angle S_{nm}$ ($n = 1, 2, 3$ and $m = 1, 2, 3$) are functions of $A_{3\text{-stub}}$ and θ_{off} as shown in Fig. 2.

Equation (4) shows that there is coupling between the ports of the network, which indicates the mutual injection will occur between magnetrons. In addition, if port 3 is connected to a matched dummy load, no reflection occurs at this port and $|S_{33}| = 0$. Then, according to (4), the reflecting voltage at each port of the network is derived as follows;

$$\begin{bmatrix} V_1^- \\ V_2^- \\ V_3^- \end{bmatrix} = \begin{bmatrix} |V_1^+||S_{11}|e^{j(\alpha_1+\theta_1)} + |V_2^+||S_{12}|e^{j(\beta+\theta_2)} \\ |V_2^+||S_{22}|e^{j(\alpha_2+\theta_2)} + |V_1^+||S_{21}|e^{j(\beta+\theta_1)} \\ |V_1^+||S_{31}|e^{j(\gamma+\theta_2)} + |V_2^+||S_{32}|e^{j(\sigma+\theta_1)} \end{bmatrix} \quad (5)$$

Fig. 2. S -parameters' amplitude and phase under different depth combinations of stub conditions.

where V_n^- and V_n^+ denote the reflecting and incident voltages at the n th port, respectively, θ_1 – θ_3 denote the phase shifts of these three voltages.

Based on (2) and our previous work [18], the phase equation of the peer-to-peer injection-locked magnetrons of the system is obtained

$$\rho_1 = K|S_{12}| + |S_{11}|\cos(\alpha_1 - \beta - \Delta\theta) \quad (6)$$

$$\rho_2 = \frac{|S_{21}|}{K} + |S_{22}|\cos(\alpha_2 - \beta + \Delta\theta) \quad (7)$$

$$\omega_1 - \omega_2 = \frac{\omega_0}{Q_{\text{ext1}}Q_{\text{ext2}}} [Q_{\text{ext1}}\rho_2 \sin(\beta - \Delta\theta) - Q_{\text{ext2}}\rho_1 \sin(\Delta\theta + \beta)] \quad (8)$$

where ω_i and ω_0 represent the instantaneous output frequency, resonant frequency and $K = V_2^+/V_1^+$ is the amplitude ratio of the two magnetrons' output power. $\Delta\theta = \theta_2 - \theta_1$ is the phase difference between the two magnetrons' outputs. $Q_{\text{ext}m}$ represents the m th magnetron external quality factor. The phase difference $\Delta\theta$ of the magnetrons is closely related to their operating states and the system's S -parameters.

Therefore, according to (2)–(8), the realization and control of the magnetron injection-locked states are achieved by altering the magnetron operating conditions and adjusting the system S -parameters with the three-stub tuner.

C. Characteristic of the Proposed System

The combining efficiency is the most important metric for a power-combining system, and microwave efficiency is usually defined as the ratio of the output power to the sum of the magnetron output powers. For the proposed system, since there is no high-loss component, we have $\eta_{\text{com}} = \eta_{\text{mw}}$. Thus, based on (5), efficiency of the system is expressed as follows:

$$\eta_{\text{com}} = \frac{[|S_{31}| + K|S_{32}|e^{j(\sigma-\gamma+\theta_2-\theta_1)}]^2}{1 + K^2} \times 100\% \quad (9)$$

where $K = V_2^+/V_1^+$ is the amplitude ratio of the two microwaves. The system efficiency is determined by the network's S -parameters and the phase and power differences between the two magnetrons.

Specifically, $|S_{32}|$, $|S_{31}|$, σ , and γ will be influenced by the tuning state of the three-stub tuner, while the phase difference $\theta_2 - \theta_1$ is determined by the peer-to-peer injection-locked state of magnetrons. It can be observed from (5) to (8) that the phase difference $\theta_2 - \theta_1$ is determined by the magnetron outputs and the S -parameters of the power-combining network. Therefore, by adjusting the susceptance jB of the three-stub tuner and the operating conditions of the magnetrons, the proposed system is theoretically capable of achieving high-efficiency coherent power combining.

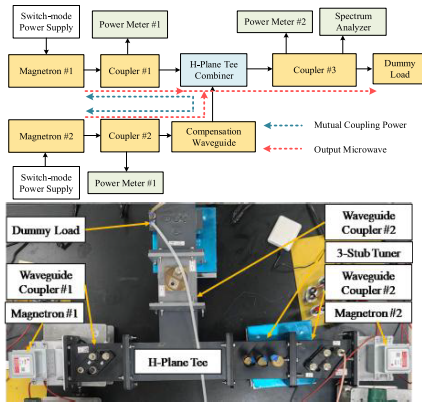


Fig. 3. Diagram and photograph of the power-combining system.

III. EXPERIMENTS AND RESULTS

A. *S*-Parameters of the System

To verify the analysis in Section II, *S*-parameters of the power-combining network shown in Fig. 1 were measured using a vector network analyzer (Agilent N5230A). The depths of the three screws were adjusted sequentially to observe how variations in screw depth affect the network's *S*-parameters.

Fig. 2 illustrates the measured magnitude and phase of the *S*-parameters of the power-combining network under different stub depths. It is observed that changing screw depths will cause significant variations in both the amplitude and phase of the *S*-parameters. The results indicate that screw depths are expected to influence the peer-to-peer locking state of the magnetrons and power-combining efficiency, as described in (8) and (9).

B. Power-Combining Experiment

A power-combining system is established to verify the proposed method. The block diagram and photograph of the magnetron experimental system are shown in Fig. 3. Two 1-kW S-band magnetrons (LG, 2M244) are powered by two improved, controllable switch-mode power supplies (WELAMP 2000F, Magmeet). The anode currents of the two magnetrons (I_1 and I_2 , respectively) can be varied from 150 to 450 mA. The filament current supply is automatically cut off when the magnetron is operating normally. The magnetron output spectrum is measured with a spectrum analyzer (FSP, R&S), and the port power of the *H*-plane tee is measured with two power meters (AV2433 and 2438CB, Ceyear). A water-cooled dummy load absorbs the microwave power output from the magnetrons.

In the experiment, the first tuning step is adjusting the insertion depths of the stubs individually and sequentially to search for the optimal. Then, when the optimal depth for one of the screws is determined, its position is fixed, and the same procedure is applied to the remaining two screws until the optimal depths of all three stubs are obtained.

C. Results and Discussion

Fig. 4(a) shows the variation in microwave efficiency during the first tuning step, which is mentioned in Section III-B. Both magnetron anode currents, I_1 and I_2 , are originally set

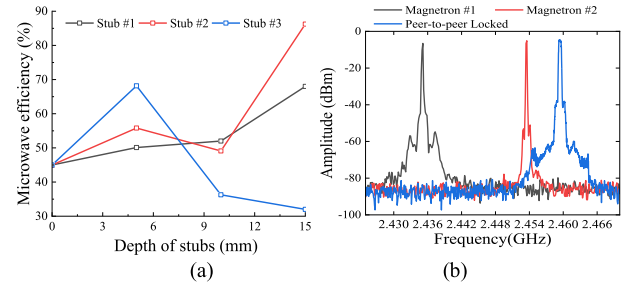


Fig. 4. (a) Microwave efficiency under various depth combination of stubs. (b) Spectra of magnetrons under best condition.

TABLE I

MEASURED MICROWAVE EFFICIENCY UNDER DIFFERENT CONDITIONS

Depth of Stubs	Magnetron#1	Magnetron#2	Output	η_{MW}
0mm, 0mm, 0mm	1050 W	1130 W	984 W	45.0%
0mm, 15mm, 0mm	848 W	1020 W	1610 W	86.2%
10mm, 15mm, 0mm	848 W	990 W	1620 W	88.1%
10mm, 16mm, 2mm	911 W	990 W	1700 W	89.4%
	674 W	924 W	1440 W	90.1%

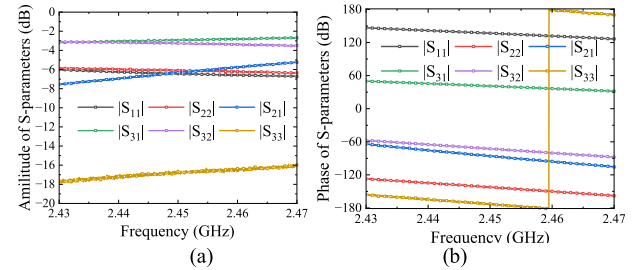


Fig. 5. (a) Amplitude and (b) phase of *S*-parameters under the best depth of the stubs.

to 400 mA. When adjusting the stub depths, the microwave efficiency varies significantly, indicating that the three-stub tuner effectively modifies the system characteristics to alter the power-combining efficiency. Among them, the adjustment of stub #2 has a particularly significant effect, with an optimal combining efficiency of 86.2%.

Then, continuing the tuning process, the optimal depths of screws were determined to be 10, 16, and 5 mm, respectively. Table I provides the detailed efficiency during the tuning process. The microwave efficiency varies with the screw depths and reaches 89.4%. Then, the output power is changed by controlling the anode current, and an optimal efficiency 90.1% is obtained. Fig. 4(b) presents the measured spectra of the two magnetrons, where the spectrum of the output signal corresponds to the max-hold spectrum for 30 s. A stable spectrum with noticeable frequency shift is observed when both magnetrons operate simultaneously, confirming the occurrence of peer-to-peer locking.

Fig. 5 shows the *S*-parameters of the system under the optimal configuration. It is observed that the *S*-parameters of the *H*-Tee undergo specific changes. This result is consistent with the theoretical analysis in Section II. The three-stub tuner modifies the system *S*-parameters, thereby influencing the combining efficiency.

The anode current of a magnetron is closely related to its output characteristics. Fig. 6 illustrates the variation in

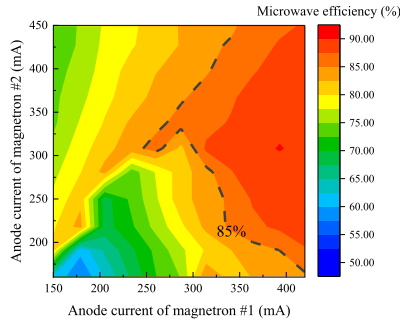


Fig. 6. Measured microwave efficiency under various anode currents.

TABLE II

MEASURED MICROWAVE EFFICIENCY UNDER DIFFERENT CONDITIONS

Work	Microwave Efficiency	Injection-Locking	Isolator	Tunability	Phase noise (dBc/Hz @500kHz)
[8]	86.7%	External	Yes	Strong	-65.0
[12]	94.5%	External	Yes	Strong	NM
[18]	90.3%/93.6% /93.6%	Peer-to-peer	No	Weak	-62.52/ 66.74/ 60.21
This work	90.1%	Peer-to-peer	No	Normal	-67.7

microwave efficiency under different anode currents of magnetrons. It can be observed that when the anode current varies, the system maintains an efficiency higher than 85% over a wide tuning range, as indicated by the dashed line in Fig. 6. In addition, a maximum microwave efficiency of 90.1% with an output power of 1440 W is achieved at anode currents of 391 and 307 mA, respectively. This result shows the proposed system is not limited to achieving high efficiency only at the tuned operating point but also exhibits a good operational stability, thereby confirming the feasibility of the proposed approach.

Table II provides a comparison between this work and the reported work. The proposed approach achieves an efficiency comparable to that reported in [8] and [18], but lower than that in [12], which employs external injection for precise phase control. However, this approach does not require isolators or phase shifters, and offers more tunability compared with [18]. This work provides a novel idea for establishing a compact, low-cost, and tunable magnetron power-combining system.

IV. CONCLUSION

A novel system for achieving high-efficiency coherent power combining of magnetrons is proposed in this letter. The injection-locking characteristic is qualitatively analyzed, and a PPL-MPC system without an isolator or phase shifter is experimentally demonstrated. The proposed system utilizes a three-stub tuner to achieve tunability and achieves a maximum microwave efficiency of 90.1%, with an output power of 1440 W. This study offers a novel approach to a high-efficiency, low-cost magnetron power-combining system and is expected to be applied to applications such as microwave chemistry, microwave heating, and MWPT.

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